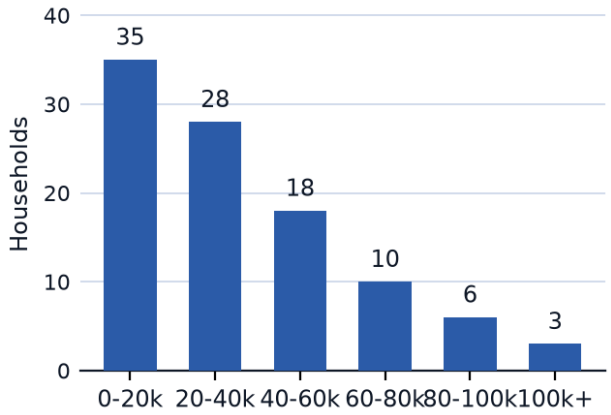
Describing Distributions & Linear Regression

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.03

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

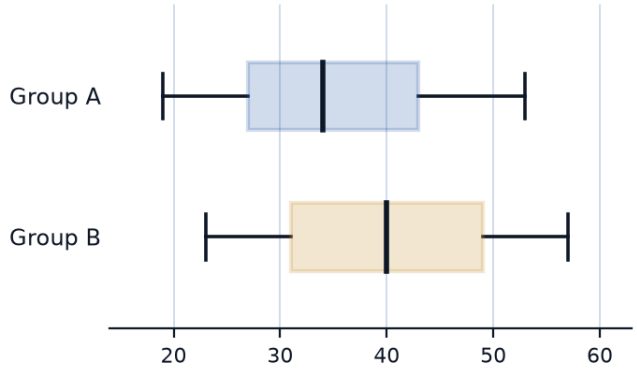
1 [Foundational] • Target: Describing shape, center, and spread

A histogram of household incomes has a long tail stretching toward high values, with most households clustered at lower incomes. Describe the shape of this distribution and state which measure of center (mean or median) is more appropriate to report, with justification.



2 [Foundational] • Target: Describing shape, center, and spread

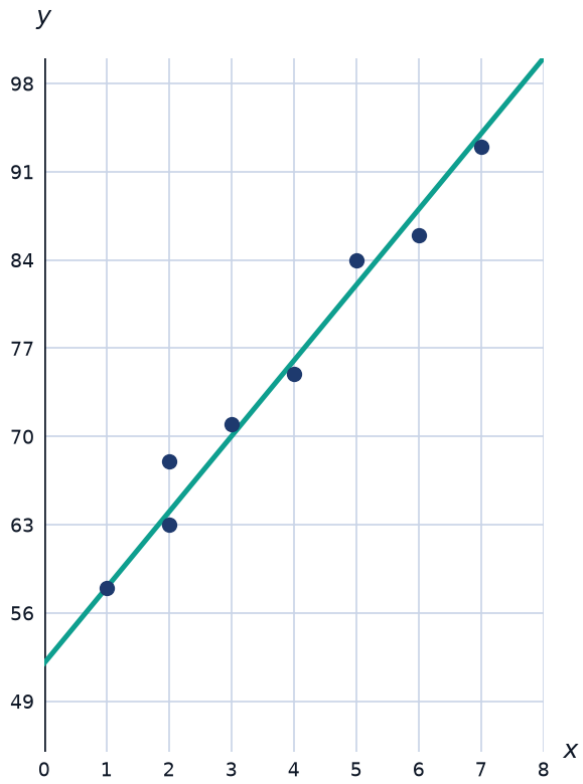
Describe what it means for a distribution to be symmetric, and state the relationship between the mean and median in that case.



© ClipSAT — clipsat.org | All Rights Reserved

3 [Foundational] • Target: Scatterplots and correlation

The scatterplot shows hours studied (x) versus test score (y) for 8 students, along with the least-squares regression line $\hat{y} = 52 + 6x$.



a Based on the scatterplot, describe the direction and strength of the association between hours studied and test score.

b Use the regression equation to predict the test score for a student who studies 4.5 hours.

4 [Foundational] • Target: Interpreting the correlation coefficient

Explain what a correlation coefficient of $r = 0.85$ indicates about a linear relationship, and explain why r alone cannot confirm that x causes y.

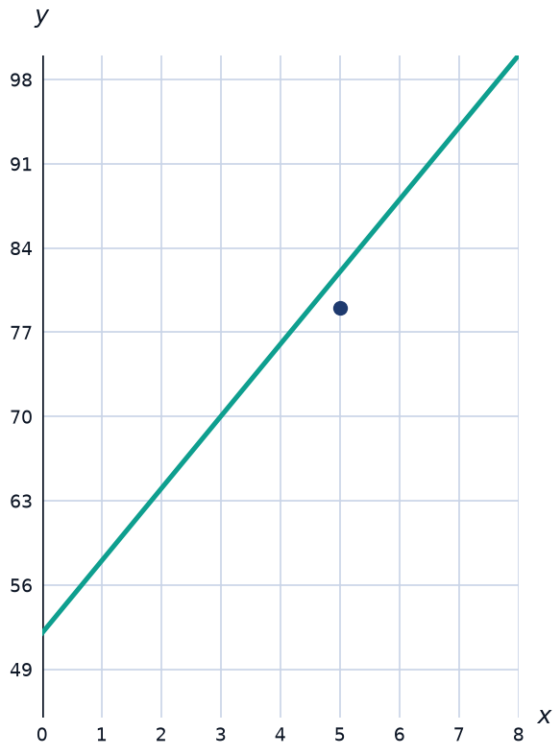
5 [Foundational] • Target: Interpreting the correlation coefficient

A data set has correlation coefficient $r = -0.4$. Describe the direction and strength of the linear relationship.

© ClipSAT — clipsat.org | All Rights Reserved

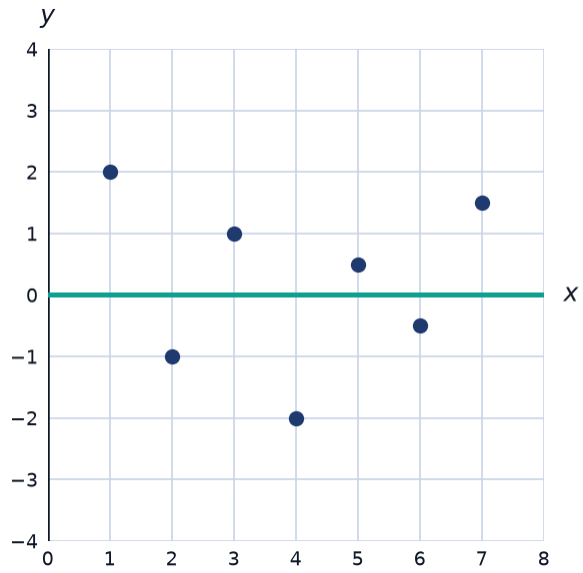
6 [Foundational] • Target: Least-squares regression and residuals

Using the regression equation $\hat{y} = 52 + 6x$ from earlier, find the residual for a student who studied 5 hours and actually scored 79.



7 [Foundational] • Target: Least-squares regression and residuals

Explain what it means for a residual plot to show "no clear pattern," and why this supports using a linear model for the data.



© ClipSAT — clipsat.org | All Rights Reserved

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Coefficient of determination**

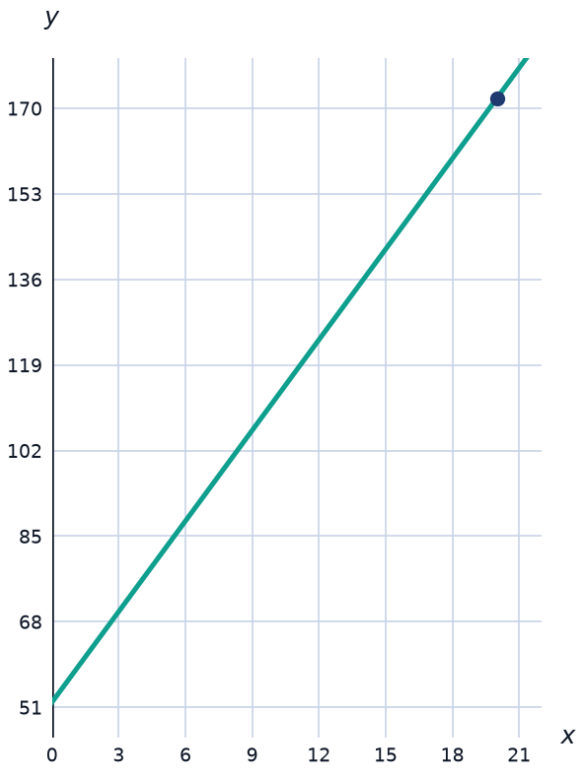
For the study-hours regression, $r = 0.85$. Find r^2 and interpret its meaning in context.

9 **[Intermediate]** • **Target: Influential points and outliers in regression**

A data point has an unusually large x-value far from the rest of the data, and removing it noticeably changes the slope of the regression line. Explain why this point is called influential, and contrast it with a point that has a large residual but does not change the line much when removed.

10 **[Intermediate]** • **Target: Extrapolation**

Using the regression $\hat{y} = 52 + 6x$ (fit using data from students who studied between 1 and 7 hours), explain why predicting a test score for a student who studied 20 hours would be unreliable.



11 **[Intermediate]** • **Target: Outliers, Leverage, and Influential Points**

A least-squares regression line predicting exam score (y) from homework completion percentage (x) is $\hat{y} = 38.0 + 0.55x$ with correlation $r = 0.82$. (a) Interpret the slope 0.55 in context. (b) Calculate and interpret the coefficient of determination r^2.

12 **[Intermediate]** • **Target: Describing shape, center, and spread**

Using the regression model $\hat{y} = 38.0 + 0.55x$, a student who completed 80% of homework scored 86 on the exam. (a) Calculate the residual. (b) Explain what a random scatter of points around 0 on a residual plot indicates about model adequacy.

© ClipSAT — clipsat.org | All Rights Reserved

13 [Intermediate] • Target: Scatterplots and correlation

A scatterplot is shown below. What is the approximate value of the correlation coefficient, r, for the data presented in the scatterplot?

14 [Intermediate] • Target: Interpreting the correlation coefficient

A researcher is designing a complex experiment to investigate the impact of three different fertilizer treatments (A, B, C) on crop yield, while also controlling for two different irrigation methods (I1, I2) and two soil types (S1, S2). To ensure a robust comparison and account for potential interactions, a stratified randomized block design is employed. Within each unique combination of irrigation method and soil type (forming the blocks), the three fertilizer treatments will be randomly assigned to distinct plots. What is the total number of unique experimental units (plots) that will be established for this experiment?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

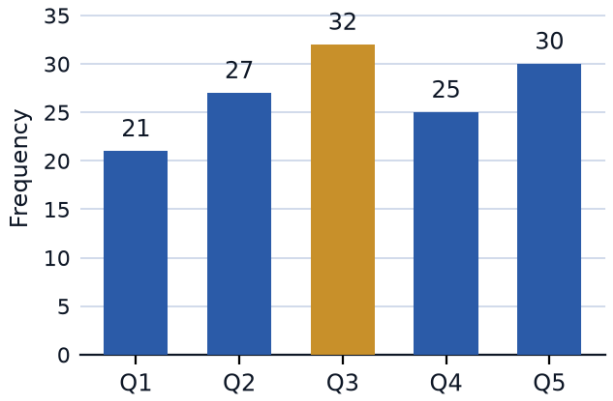
15 [Challenging] • Target: Least-squares regression and residuals • Cross-Lesson: 3.03 The Normal Distribution

A cereal manufacturer places a collectible card in each box: 20% of boxes have Card A, 30% have Card B, and 50% have Card C. Design a simulation using random digits to estimate the probability that a customer must buy at least 6 boxes to collect all three cards.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Coefficient of determination • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A researcher is designing an experiment to test the effectiveness of a new fertilizer on plant growth. They have a total of 50 experimental plots available. The experiment will involve two treatments: the new fertilizer (Treatment A) and a standard fertilizer (Treatment B). To ensure fair comparison and control for confounding variables, they decide to use a randomized block design. The plots are divided into 5 blocks based on soil quality, with each block containing 10 plots. Within each block, the plots will be randomly assigned to either Treatment A or Treatment B. If the number of plots assigned to Treatment A within the first block is x, and the total number of plots assigned to Treatment A across all 5 blocks is 25, and the proportion of plots assigned to Treatment A in block 1 is $\frac{1}{5}$ of the total plots in block 1, what is the value of x?



17 [Challenging] • Target: Influential points and outliers in regression • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

The commute times for employees at a large tech company have a strongly right-skewed distribution with mean $\mu = 38$ minutes and standard deviation $\sigma = 15$ minutes. A random sample of 49 employees is selected. (a) Describe the shape, mean, and standard error of the sampling distribution of the sample mean $\bar{x}$. (b) Calculate $P(\bar{x} > 42 \text{ minutes})$.

18 [Challenging] • Target: Extrapolation • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A data set consists of the following five points:

$P_1 = (2, 3), P_2 = (4, y), P_3 = (6, 7), P_4 = (8, 9), P_5 = (10, 11)$. If the mean of the x-coordinates is 6 and the mean of the y-coordinates is 7, what is the value of y?

19 [Challenging] • Target: Interpreting the Least-Squares Regression Line (LSRL) • Cross-Lesson: 1.01 Measures of Center & Spread

A researcher is studying the relationship between study hours and final exam scores for a group of AP Statistics students. The researcher collected data and constructed a scatterplot to visualize the relationship. The scatterplot includes a line of best fit. The equation of the line of best fit is given by $\hat{y} = 2.5x + 15$, where x represents the number of hours studied per week, and $\hat{y}$ represents the predicted final exam score. If a student studied for 12 hours, what is the predicted final exam score for that student?

© ClipSAT — clipsat.org | All Rights Reserved

20 [Challenging] • Target: Residuals and Residual Plots • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

The following diagram shows a scatterplot comparing the number of hours studied per week for a math course and the final exam score for a sample of students. What is the approximate y-intercept of the line of best fit?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Describing shape, center, and spread

A least-squares regression line is $\hat{y} = 16 - 5x$. Predict $\hat{y}$ when $x = 12$.

a. 76

b. -46

c. -42

d. -44

22 [Foundational] • Target: Scatterplots and correlation

A least-squares regression line is $\hat{y} = 82 + 3x$. Predict $\hat{y}$ when $x = 1$.

a. 87

b. 83

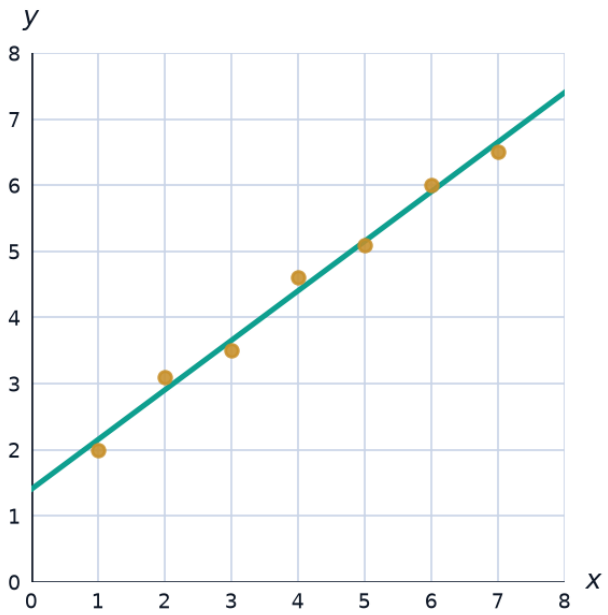
c. 79

d. 85

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Interpreting the correlation coefficient

A least-squares regression line is $\hat{y} = 11 + 8x$. Predict $\hat{y}$ when $x = 5$.



a. -29

b. 49

c. 53

d. 51

24 [Foundational] • Target: Least-squares regression and residuals

A least-squares regression line is $\hat{y} = 55 + 3x$. Predict $\hat{y}$ when $x = 13$.

a. 16

b. 96

c. 94

d. 92

25 [Foundational] • Target: Coefficient of determination

A least-squares regression line is $\hat{y} = 32 - 4x$. Predict $\hat{y}$ when $x = 15$.

a. -26

b. -30

c. -28

d. 92

26 [Foundational] • Target: Influential points and outliers in regression

A least-squares regression line is $\hat{y} = 39 - 5x$. Predict $\hat{y}$ when $x = 18$.

a. -51

b. -49

c. 129

d. -53

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Extrapolation

A least-squares regression line is $\hat{y} = 79 + 4x$. Predict $\hat{y}$ when $x = 1$.

a. 75

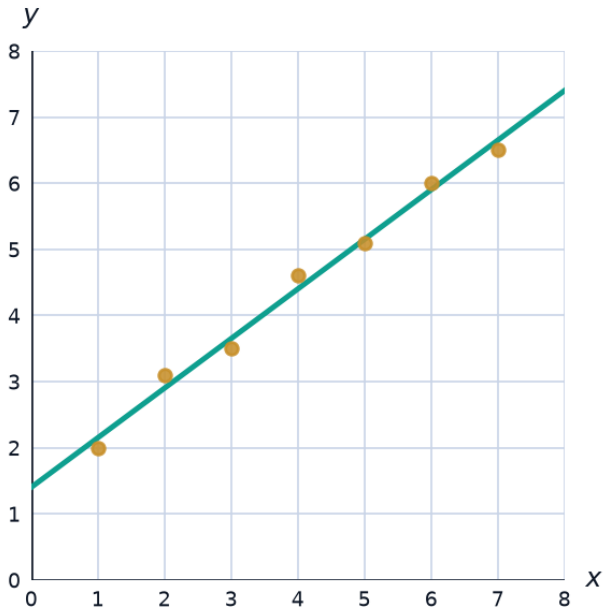
b. 83

c. 85

d. 81

28 [Foundational] • Target: Interpreting the Least-Squares Regression Line (LSRL)

A least-squares regression line is $\hat{y} = 52 + 2x$. For a point with $x = 2$ and observed value $y = 49$, find the residual (observed − predicted).



a. -9

b. -5

c. 7

d. -7

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Residuals and Residual Plots

A least-squares regression line is $\hat{y} = 73 - 9x$. For a point with $x = 6$ and observed value $y = 29$, find the residual (observed − predicted).

a. 8

b. 12

c. 10

d. -10

30 [Intermediate] • Target: Correlation Coefficient r and Coefficient of Determination r^2

A least-squares regression line is $\hat{y} = 36 - 3x$. For a point with $x = 13$ and observed value $y = 1$, find the residual (observed − predicted).

a. 2

b. 4

c. -4

d. 6

31 [Intermediate] • Target: Outliers, Leverage, and Influential Points

A least-squares regression line is $\hat{y} = 26 + 7x$. For a point with $x = 2$ and observed value $y = 41$, find the residual (observed − predicted).

a. 3

b. 1

c. 4

d. -1

32 [Intermediate] • Target: Describing shape, center, and spread

A least-squares regression line is $\hat{y} = 21 - 2x$. For a point with $x = 10$ and observed value $y = -4$, find the residual (observed − predicted).

a. -5

b. -3

c. 5

d. -7

33 [Intermediate] • Target: Scatterplots and correlation

A least-squares regression line is $\hat{y} = 64 + 7x$. For a point with $x = 5$ and observed value $y = 101$, find the residual (observed − predicted).

a. 4

b. 0

c. 2

d. -2

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Interpreting the correlation coefficient

A least-squares regression line is $\hat{y} = 45 - 9x$. For a point with $x = 15$ and observed value $y = -91$, find the residual (observed − predicted).

a. -1

b. 1

c. -4

d. -3

35 [Intermediate] • Target: Least-squares regression and residuals

A regression has correlation coefficient $r = -0.65$. Find r^2, expressed as a percentage, to two decimal places.

a. 42.25%

b. 44.25%

c. 40.25%

d. 65.00%

36 [Intermediate] • Target: Coefficient of determination

A regression has correlation coefficient $r = 0.50$. Find r^2, expressed as a percentage, to two decimal places.

a. 50.00%

b. 25.00%

c. 27.00%

d. 23.00%

37 [Intermediate] • Target: Influential points and outliers in regression

A distribution has mean 94 and median 82. Based on this comparison, classify the shape of the distribution.

a. Approximately symmetric (mean and median are close)

b. Skewed right (mean noticeably greater than median)

c. Skewed left (mean noticeably less than median)

d. Cannot be determined from the mean and median alone

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Extrapolation • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A distribution has mean 83 and median 84. Based on this comparison, classify the shape of the distribution.

a. Approximately symmetric (mean and median are close)

b. Skewed right (mean noticeably greater than median)

c. Skewed left (mean noticeably less than median)

d. Cannot be determined from the mean and median alone

39 [Challenging] • Target: Interpreting the Least-Squares Regression Line (LSRL) • Cross-Lesson: 1.01 Measures of Center & Spread

A regression has coefficient of determination $r^2 = 0.64$. Find the possible value(s) of the correlation coefficient r.

a. ± 0.64

b. 0.80

c. ± 0.90

d. ± 0.80

40 [Challenging] • Target: Residuals and Residual Plots • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A regression has coefficient of determination $r^2 = 0.16$. Find the possible value(s) of the correlation coefficient r.

a. ± 0.16

b. ± 0.50

c. 0.40

d. ± 0.40

41 [Challenging] • Target: Interpreting the Least-Squares Regression Line (LSRL) • Cross-Lesson: 2.01 Sampling Methods & Bias

A regression model predicting used car price in dollars from age in years is $\hat{y} = 24000 - 1800x$. Interpret the slope -1800 in context.

a. For each additional year of car age, the predicted price decreases by 1,800 dollars

b. A car loses exactly 1,800 dollars every year

c. When the car is 0 years old, its predicted price is $-1,800$ dollars

d. The correlation between age and price is -1800

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Interpreting the Least-Squares Regression Line (LSRL) • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

Why is the line called the 'least-squares' regression line?

a. It minimizes the sum of the squared vertical residuals $\sum(y - \hat{y})^2$

b. It minimizes the sum of squared distances to the origin

c. It maximizes the square of the correlation coefficient

d. It guarantees all residuals are perfect squares

43 [Challenging] • Target: Residuals and Residual Plots • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

Which pattern in a residual plot indicates that a linear model is appropriate?

a. A random scatter of points with no discernible pattern around the residual $= 0$ line

b. A clear curved U-shape pattern

c. A fan shape where residuals expand as x increases

d. Points clustered strictly along an inclined line

44 [Challenging] • Target: Correlation Coefficient r and Coefficient of Determination r^2 • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

What is the correct statistical interpretation of $r^2 = 0.75$?

a. 75% of the variation in the response variable y is explained by the linear relationship with x

b. 75% of the data points lie directly on the regression line

c. The slope of the regression line is 0.75

d. There is a 75% probability that x causes y

45 [Challenging] • Target: Outliers, Leverage, and Influential Points • Cross-Lesson: 3.03 The Normal Distribution

A point that has an extreme x-value relative to the rest of the data is said to have:

a. High leverage

b. A large residual

c. High correlation

d. Negative slope

© ClipSAT — clipsat.org | All Rights Reserved